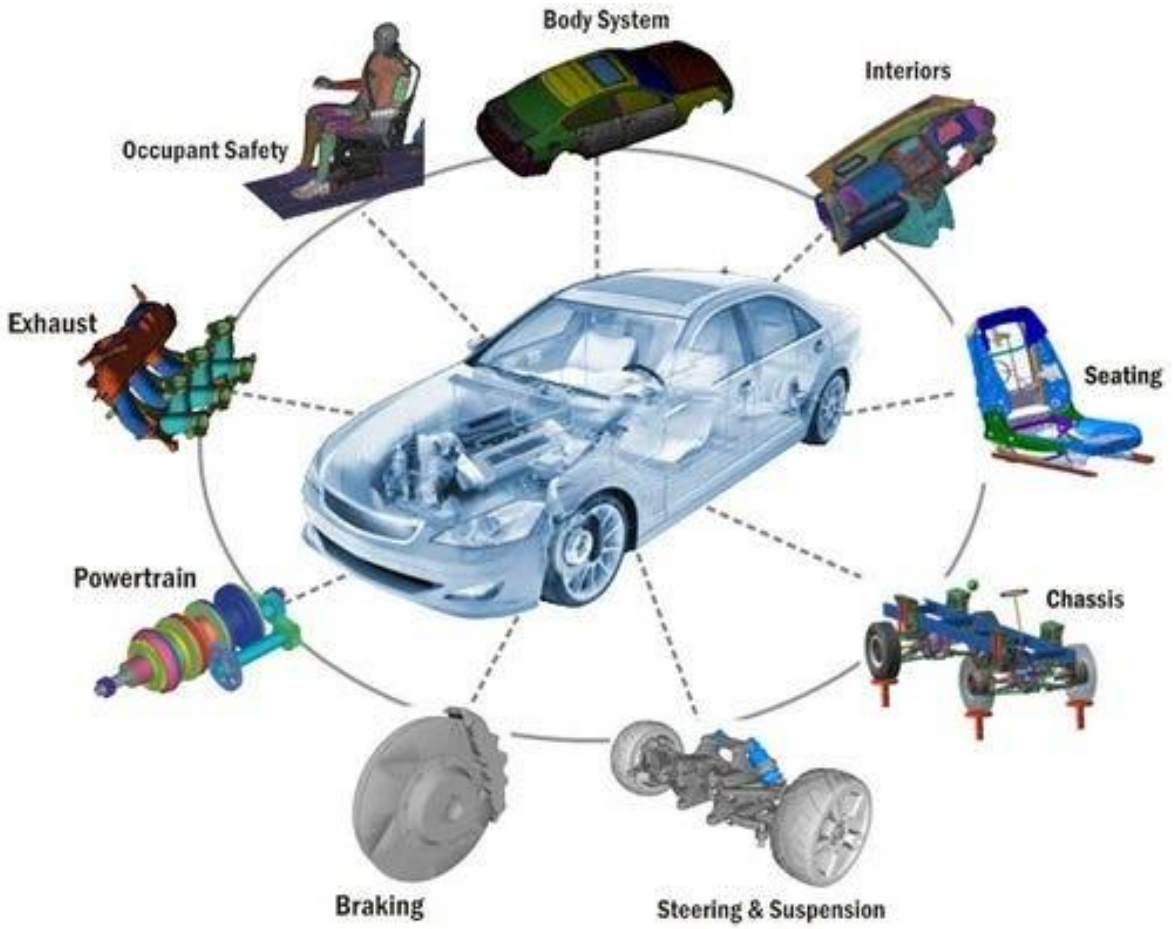
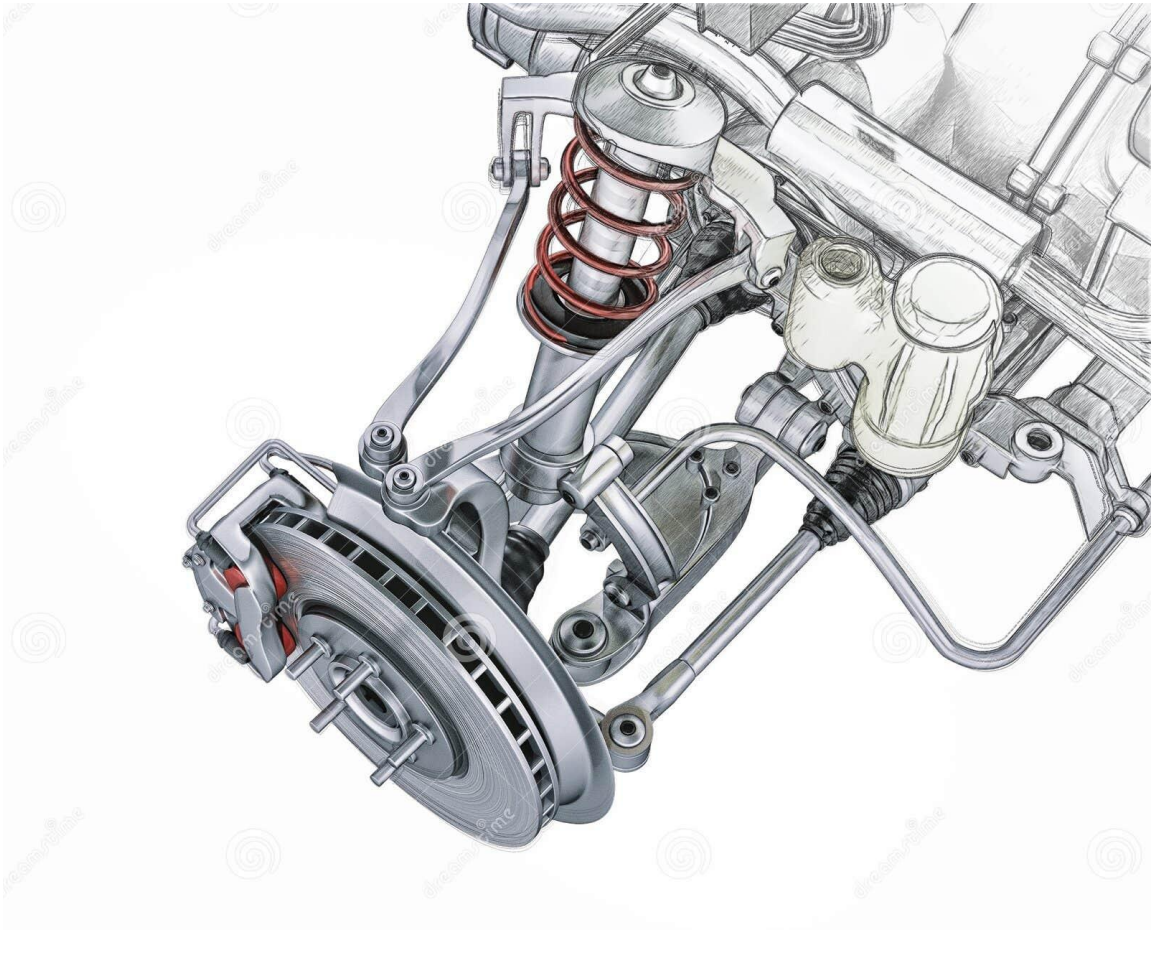




Automotive Engineering (MAM 336)

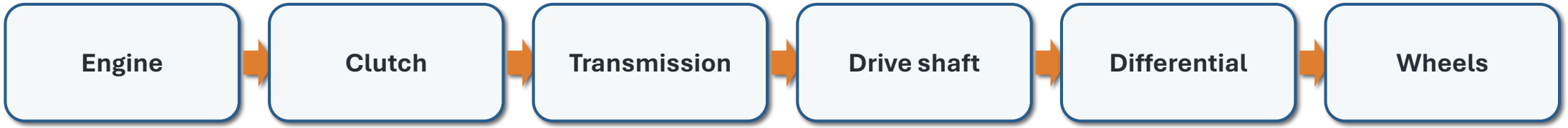
Week 3

Clutch Unit

DR/ MOHAMED EL TAHER MOHAMED

Power Flow Through the Vehicle

The drive train indicates the power flow from the engine to the driving wheels.



Power source

Torque / speed conversion

Torque distribution

Introduction

Clutch is the first part of the transmission system. It is mounted between the engine and the gear box. Clutch is a mechanism that helps to rotate the rotation from a shaft to another shaft on the same axis.

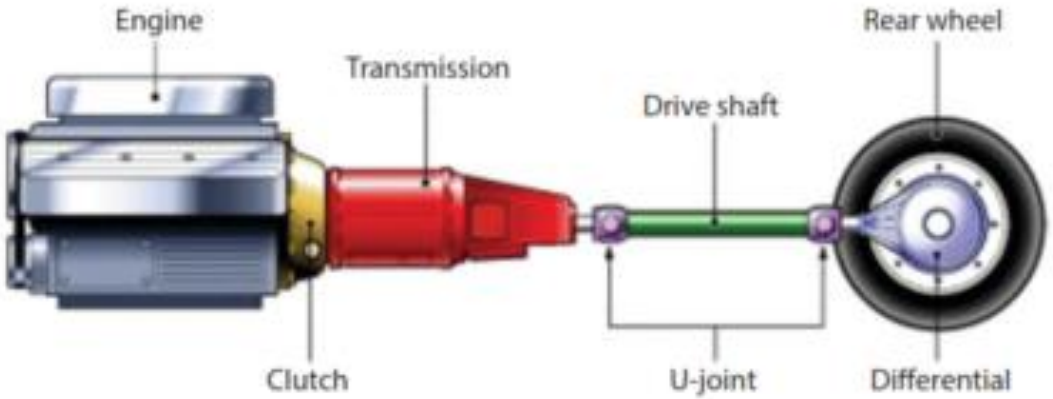


Figure 1.7 Transmission system.

Functions of the Clutch

- When the power from the engine is needed, it will connect and disconnect the gear box.
- When the Engine is not working, it will help to start the vehicle by pushing.

- Used to change gears softly, to stop the vehicle and for Engine idle.

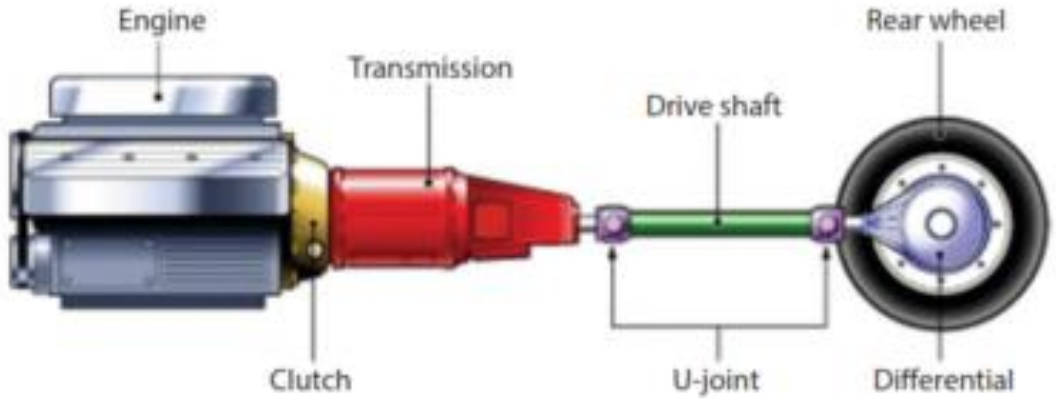
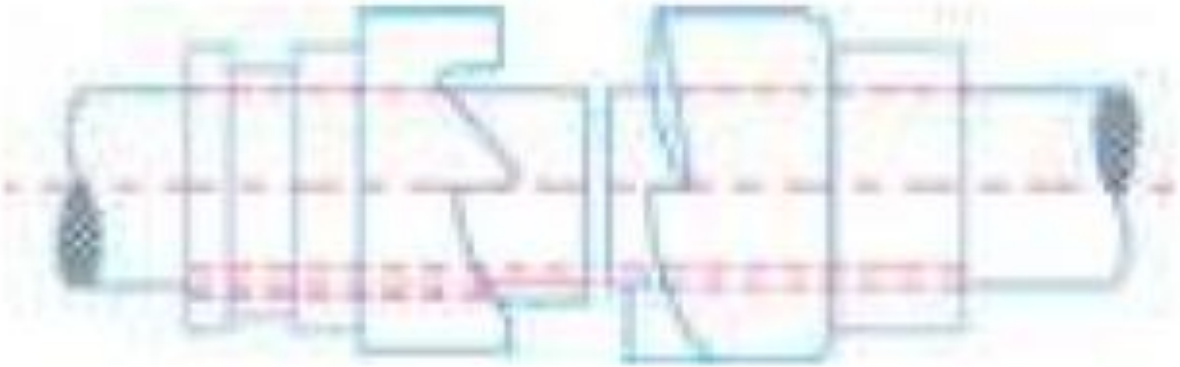


Figure 1.7 Transmission system.

Principle of the Clutch

Clutch works on the principle of friction. When the rotating disk C connects with disk D, both will start to rotate. This is the basic principle in Clutch.



Qualities of Good Clutch

1. Should have high torque transmission.
2. Should join in step-by-step.
3. Should be a good transfer of heat.
4. Operating balance is required.
5. Should be able to bear the shocks.
6. Should have free pedal play.
7. Should be easy to operate.
8. Should be in simple construction, cheap in cost and high working life.
9. Low space should be adequate.
10. Should have high co-efficient of friction.

1. Clutch Plate
2. Clutch Lining
3. Pressure Plate
4. Coil Spring
5. Release bearing
6. Clutch linkage



Clutch Plate

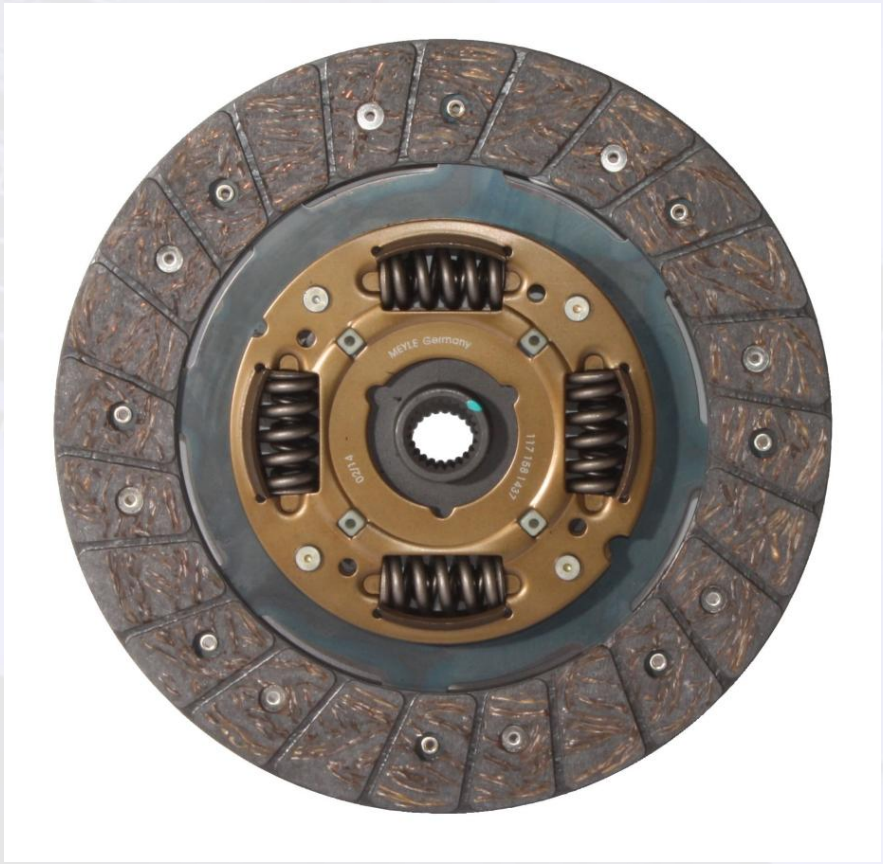
Clutch plate should be made of steel. Hub is situated in centre of the clutch plate. Hub consists of splines. Clutch plate has moving capacity. Hub is attached with disk. Disk is surrounded by many cushion springs. Above this, friction lining is attached with rivet or glue. Helical spring is attached in centre of the plate to bear torque. This is called as torsional or damper spring.



Clutch Lining

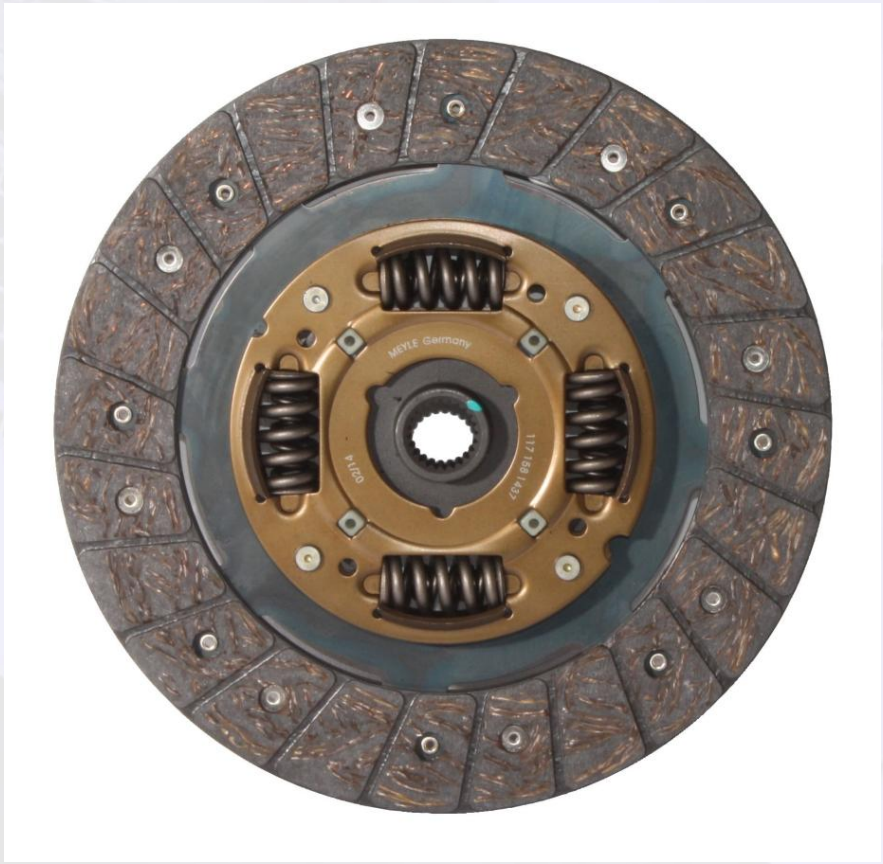
The two ends of the clutch lining and clutch plate are attached with the help of rivets or some special glue. The clutch lining is made up of following materials.

1. Asbestos
2. Reybestos
3. Fibre
4. Leather



Clutch Spring

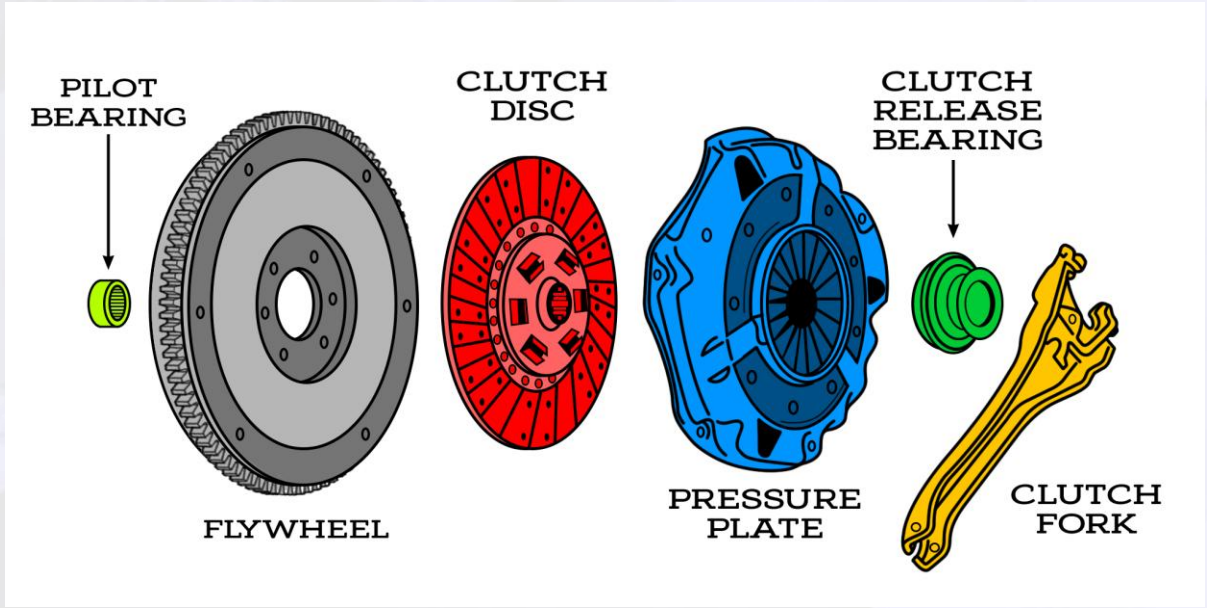
Cushion or Torsion spring is used to bear frictions produced from clutch engage and disengage.



Release Bearing and Clutch linkage

Clutch housing inside, without any contact with clutch shaft it will move forward and backward. It will press the fingers equally so it is known as Release bearing.

Operates release bearing from the clutch pedal with the help of fork is known as clutch linkage. In car it works with the cables and in heavy vehicles, it works with the steel rods. It is designed in such a way that it can able to change its length.



Types of Clutch

1. Single plate clutch
2. Multi plate clutch
3. Diaphragm clutch
4. Centrifugal clutch
5. Hydraulic clutch
6. Fluid flywheel clutch
7. Electric clutch

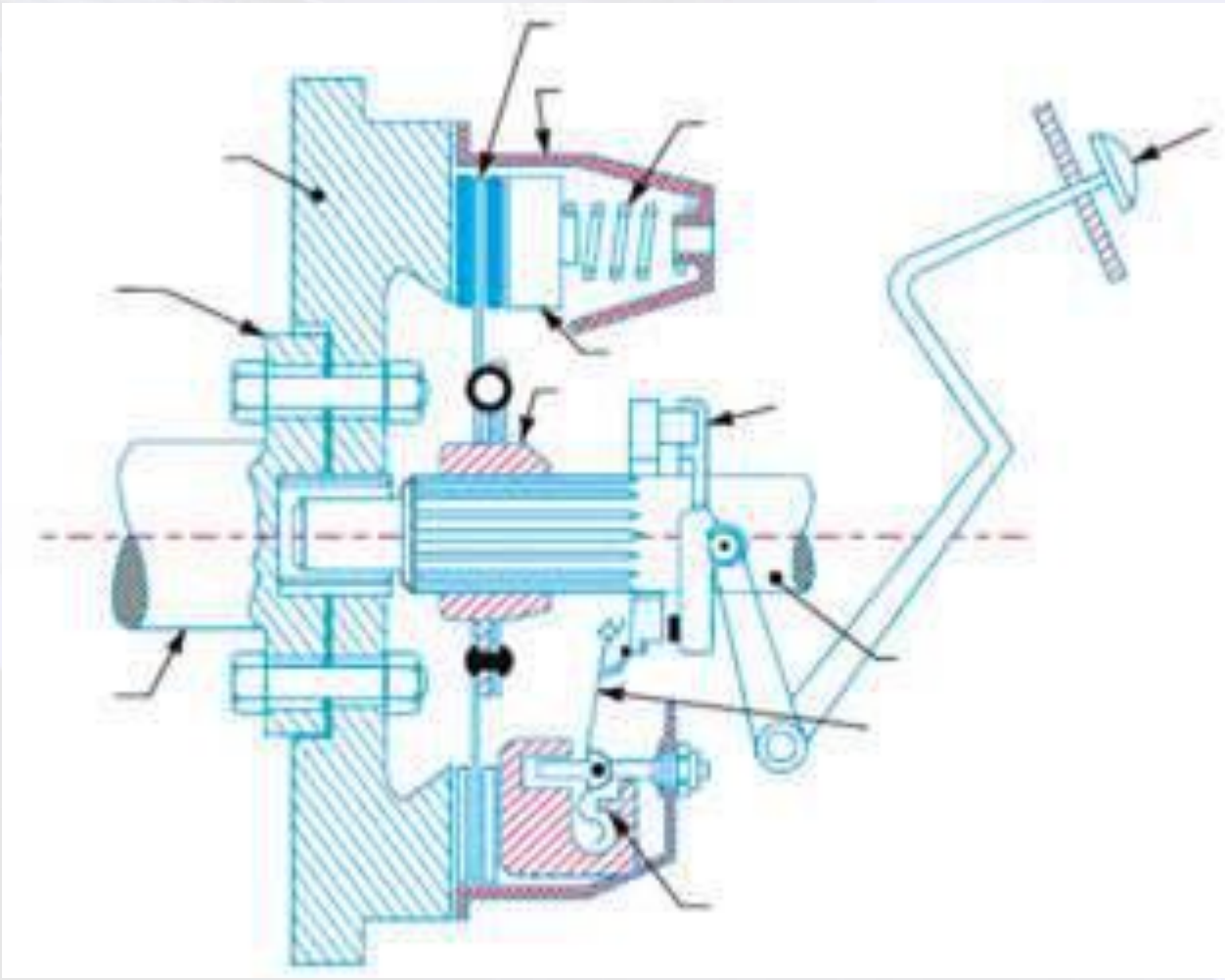
Single Disc or Plate Clutch

1. Fly wheel
2. Clutch plate
3. Clutch lining
4. Pressure plate
5. Coil spring
6. Release lever
7. Clutch shaft
8. Release lever bearing

- Clutch plate is placed in between fly wheel and pressure plate.
- Centre of clutch plate is attached with clutch shaft.
- Coil spring is placed in between pressure plate and clutch cover.
- Release lever is attached with pressure plate.
- Clutch cover is attached with fly wheel with the help of bolt and nuts.

Single Disc or Plate Clutch





Single Disc or Plate Clutch

Working principle

When clutch pedal is pressed

- When clutch pedal is pressed, release bearing will push the release lever.
- So pressure plate will move backward in opposite to spring force.
- In this stage, a gap will form in between clutch plate and pressure plate.
- Therefore, power from the engine is not transmitted to gear box.
- This is called as clutch disengaged.

When clutch pedal is released

- Pressure plate is pressed by coil spring force.
- So pressure plate connects the clutch plate with the fly wheel.
- Now, power from the engine is transmitted to gear box with the help of clutch shaft.
- Therefore, this is called as clutch engaged.

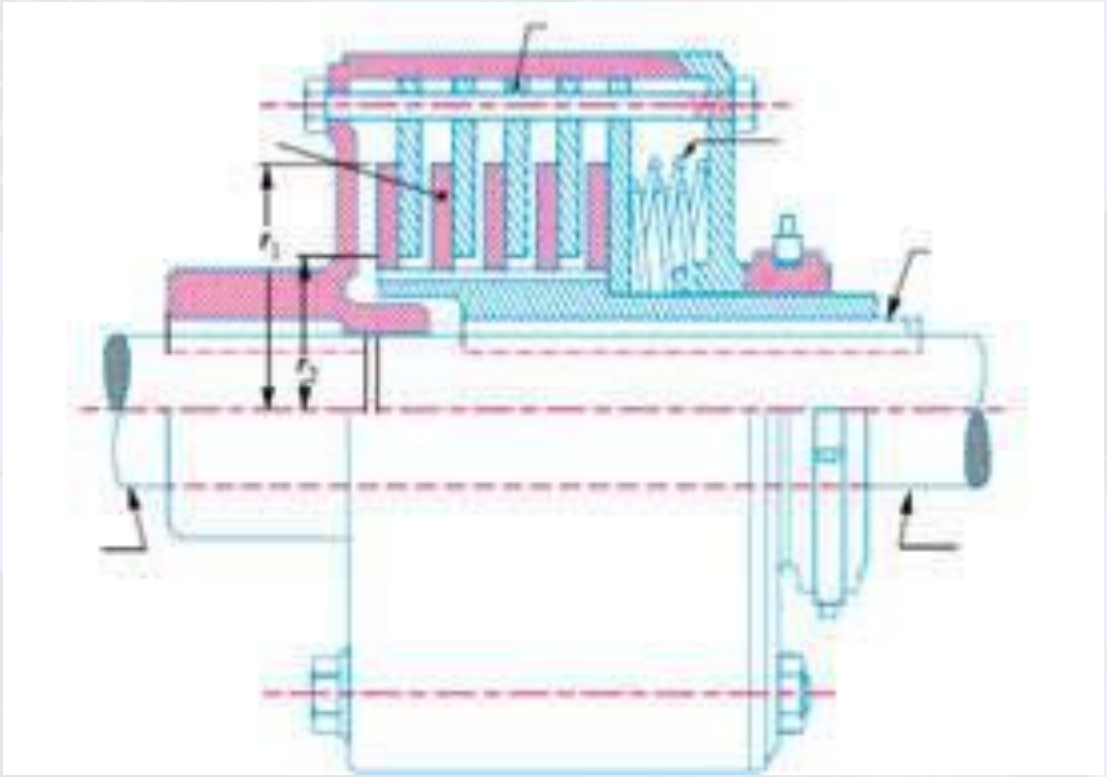
Multi plate clutch

Working principle

When clutch pedal is pressed

- Release lever is pressed by release bearing.

- So pressure plate is moving backward opposite to coil spring pressure.
- At this stage, a gap will form in between clutch plate and pressure plate.
- So the power from engine is not transmitted to gear box.
- This is called as clutch disengaged.



Multi plate clutch

When clutch pedal released

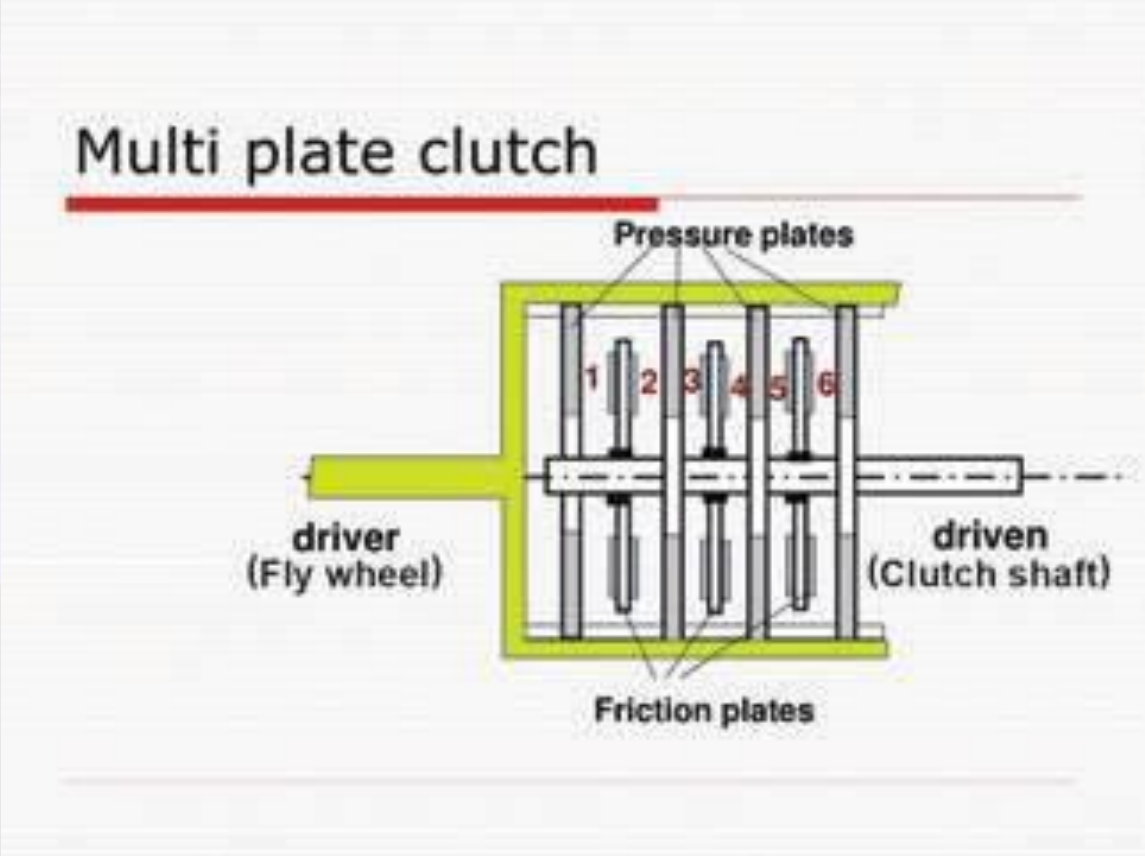
- Pressure plate is pressed by coil spring force.
- So pressure plate connects the clutch plate with fly wheel.
- Now, engine power is transmitted to gear box.
- This is called as clutch engaged.

Advantages

1. Transmits more torque.
2. Smooth joining
3. Less noise
4. More life.

Disadvantages

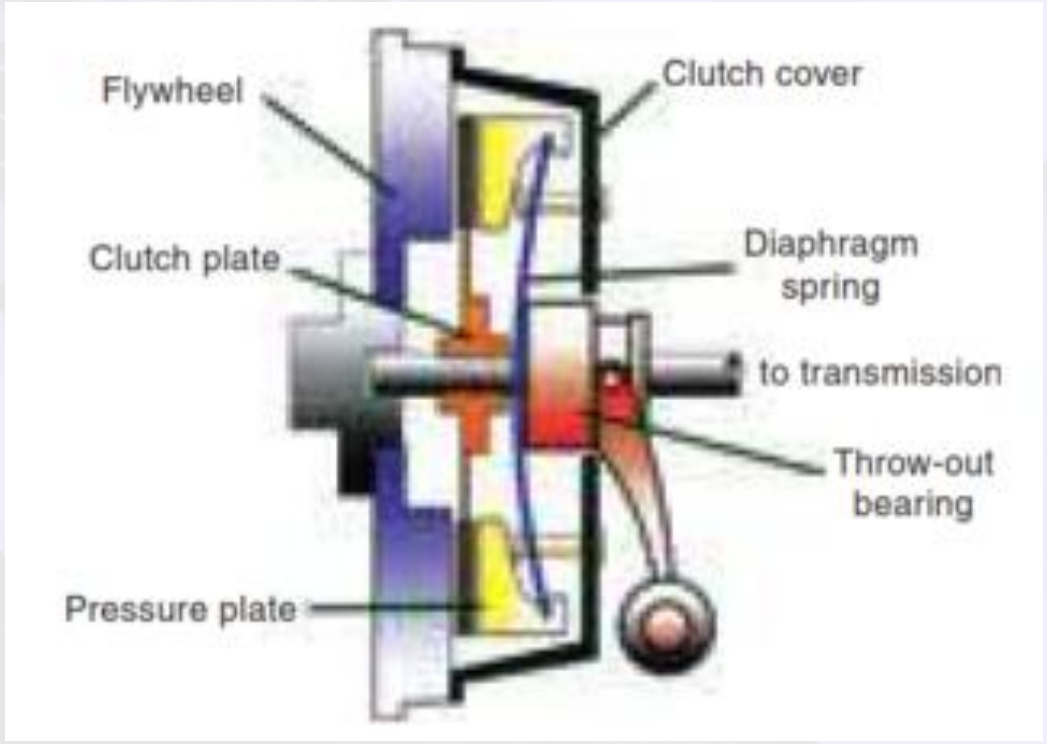
1. Not used in heavy vehicles.
2. Difficult in maintenance
3. High initial cost



Diaphragm clutch

Construction

- Here instead of coil spring, diaphragm spring is used.
- Clutch plate is placed in between fly wheel and pressure plate.
- The centre of clutch plate is attached with clutch shaft.
- Outside end of the diaphragm spring is attached with with pressure plate by clamp, screw, and clutch cover with two pivot rings.



Diaphragm clutch

Working principle

When clutch pedal is pressed

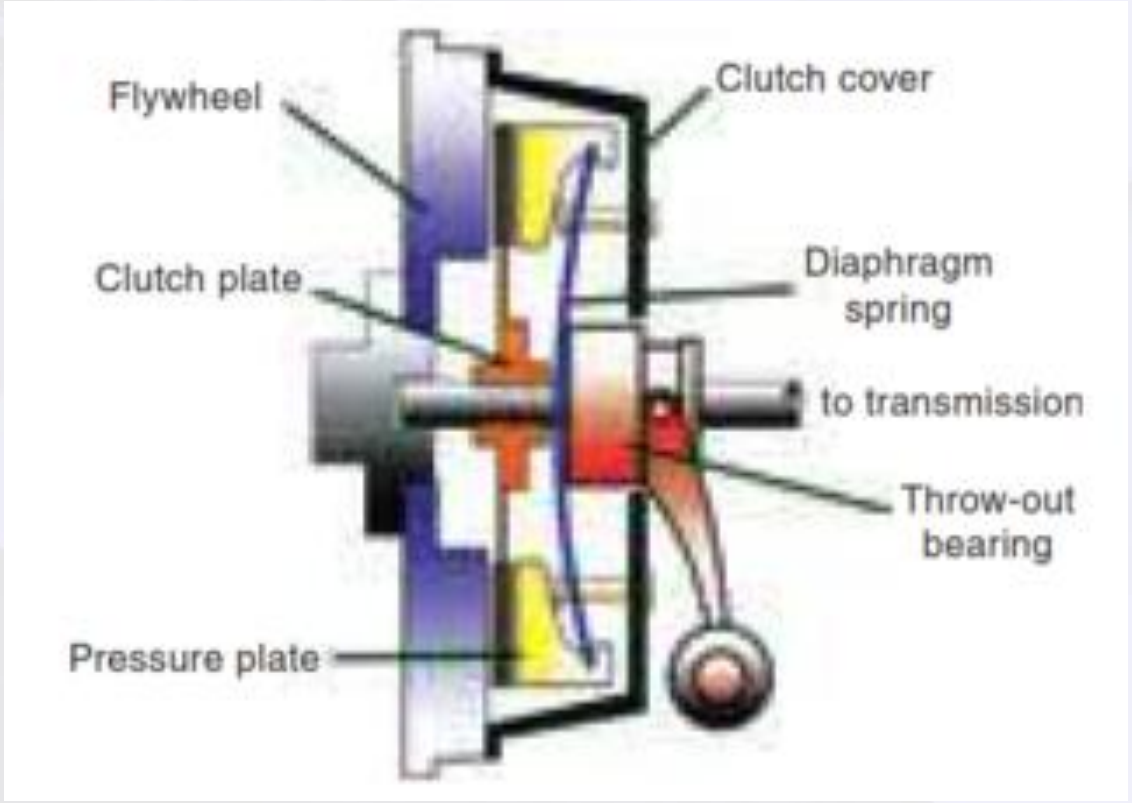
- Thrust bearing is moving forward and presses the centre of diaphragm spring.
- So outside end of the diaphragm spring pulls pressure plate in backward direction.
- Now, a gap is formed in between pressure plate and clutch plate.
- Therefore, a engine power is not transmitted to gear box.
- This is called as Clutch disengaged.

When clutch pedal is released

- Outside end of the diaphragm spring pushes the pressure plate.
- So clutch plate joins with fly wheel.
- Therefore, engine power is transmitted to gear box.

Advantages

1. Transmits more torque.
2. Low maintenance is required.
3. Low space is required.
4. More life.
5. Works smoothly with less noise.



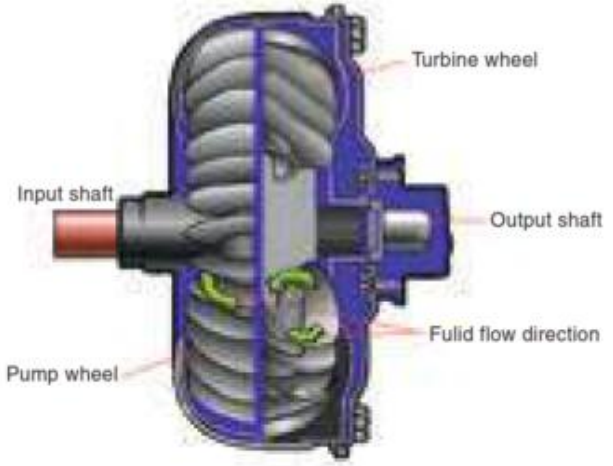
Fluid fly wheel

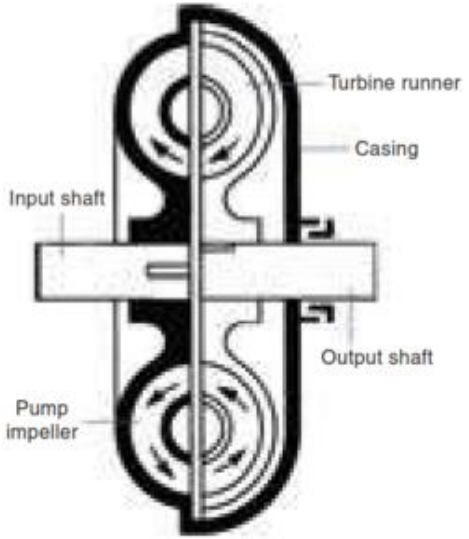
Components

1. Crank shaft
2. Gear box shaft
3. Pump
4. Turbine
5. Vanes
6. Oil seal
7. Oil

Construction

- Pump is attached with fly wheel, which operates by a component called Driving member.
- Turbine is attached with gear box shaft. This is called as Driven member.
- Inside of pump and turbines, vanes are there. Vanes are filled with oil at a particular level.
- Oil seal is attached to stop the oil leakage.





Fluid fly wheel

Operating method

At Idle speed

- When engine rotates with this speed, pump attached with crank shaft operates easily.
- Now due to less centrifugal force, oil will not move fast towards turbine.
- Therefore, turbine will not rotate and engine power is not transmitted to gear box.

At High speed

- When engine rotates with this speed, pump rotates fastly.
- Now due to high centrifugal force, oil will move fast towards turbine blade.
- Therefore, turbine rotates fastly and engine power is transmitted to gear box.

Advantages

1. Engine power transmission takes place without vibration.
2. Easy maintenance.
3. No need of adjustment.
4. No need of clutch pedal to operate this.
5. Simple construction.
6. Works smoothly without noise.
7. More life.
8. Gears cane be changed easily.

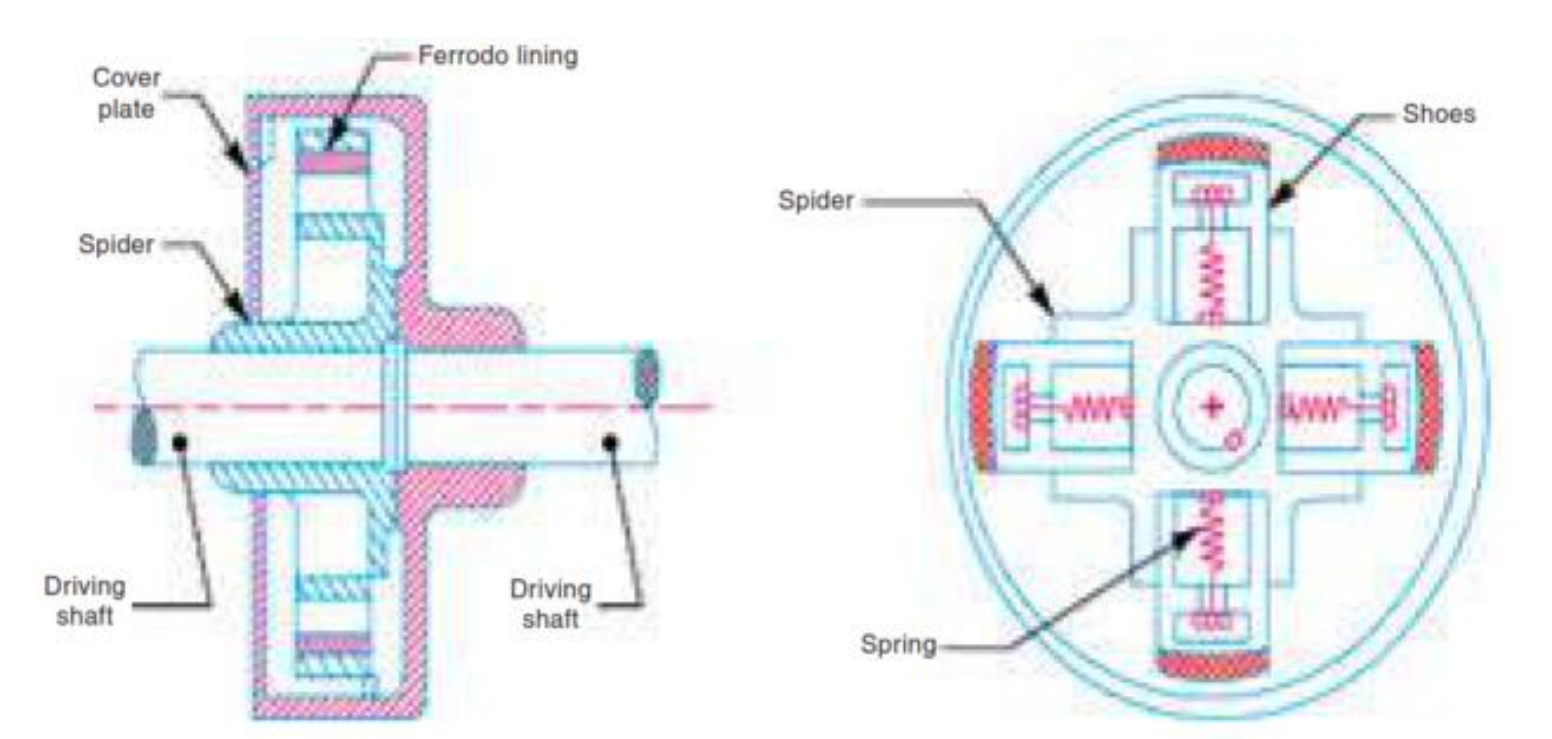
Disadvantages

1. In all speed, up to 2% power loss will be there.
2. Clutch will not work properly when leakage of oil.
3. It is suitable for only automatic gear box.

Centrifugal clutch

Construction

- It works under centrifugal force. So it is known as centrifugal clutch.
- Here, pressure plate is attached with Ball weights and Bell crank lever.
- Pressure spring is placed in between clutch plate and pressure plate. Pressure release springs are placed in between fly wheel and clutch plate.
- In this type, there is no clutch pedal.





Thank You!